Materials Engineer

Pantex Plant · Amarillo, TX

Job ID (this archive)	MFG-068
Employer	Pantex Plant
Location	Amarillo, TX
Career stage	Experienced
Location group	Texas (outside DFW)
Major (archive tag)	Materials Science & Engineering
Sector area	Quality, materials & process engineering
Industry (LinkedIn)	Defense and Space Manufacturing
Pay listed	Not listed
Stated rules	GPA 3.0+ · U.S. citizen / ITAR · Security clearance
Posted / closes	2026-10-26 · 2026-11-25
Status when checked	Accepting applications on LinkedIn (Nov 1, 2026)
Source page	https://www.linkedin.com/jobs/view/4483536627/
Captured	2026-11-01

Full posting text as it appeared on LinkedIn on Nov 1, 2026, saved for ECS 1100. Postings change and close often; open the source page above for the current version before applying.

At a glance

What you would do

Materials engineer verifying adhesives, foams and coatings used in weapon assembly, troubleshooting material failures on the line and overseeing lab tests.

Who can apply

- **Majors named:** Any engineering degree, Any STEM degree, Information Technology
- **Degree and experience:** Bachelor's or master's in engineering, science or IT; 3.0 GPA preferred; DOE O 426.2A experience; Q clearance and U.S. citizenship

Skills mentioned

Teamwork & collaboration · Communication · Problem solving & root cause · Safety · Supply chain, planning & logistics · Excel · ERP / SAP / MES · Quality tools

Full posting text

Location: Amarillo, TX - Pantex Plant

Job Title: Materials Engineer

Career Level From: Associate

Career Level To: Senior Associate

Organization: Weapon Material Program (51395467)

Job Specialty: Materials Engineering

What You'll Do

Daily Duties

Systems, Applications, and Products in Data Processing (SAP) Material Definition and Quality Verification: Spend a portion of each day working in SAP to ensure that material definitions, part numbers, raw material requisitions, and testing parameters correctly reflect the requirements dictated by specifications and drawings. This involves reviewing material lots for production and verifying that materials meet chemical and mechanical benchmarks.

Programmatic Procedure and Drawing Reviews: Systematically evaluate step-by-step weapon assembly and disassembly procedures to confirm that material call-outs (such as adhesives, sealants, foams, and coatings) strictly adhere to design requirements. Ensure that curing times, mixing ratios, and application techniques comply with the authorized engineering specifications.

Floor-level Troubleshooting and Failure Analysis: Act as the primary technical responder when process issues or material failures occur on the production line (e.g., an adhesive fails to cure, a coating exhibits poor adhesion, or a polymer shows unexpected cracking) or during acceptance testing. You will initiate root-cause investigations, conduct engineering evaluations, and coordinate with Quality Assurance to document the issue via Non-Conformance Reports (NCRs), generating the technical justifications required for "use-as-is" or scrap decisions.

Laboratory Testing Oversight: Monitor active laboratory testing or pilot-scale blending and packaging operations. Ensure that the daily execution of material testing procedures (such as tensile strength, durometer hardness, thermal properties, or chemical composition) aligns with testing guidance, calibrating equipment as required.

Weekly & Collaborative Duties

Process Engineering Collaboration: Meet regularly with Pantex Process Engineers to align material processing requirements with the physical realities of the production bay. This includes confirming that material working-life constraints fit inside the weapon assembly schedules and that specialized tooling accommodates material behaviors (like viscosity and thermal expansion).

Design Agency and Production Agency (PA) Coordination: Maintain continuous technical contact with Design Agency counterparts at Los Alamos National Laboratory (LANL), Sandia National Laboratories (SNL), Lawrence Livermore National Laboratory (LLNL), and other relevant system sites, such as the Kansas City National Security Campus (KCNSC). This ensures that Pantex’s manufacturing plans remain in perfect synchronization with design intent and weapon certification requirements.

At-Risk Material Monitoring and Supply Chain Mitigation: Proactively identify raw materials, chemical precursors, or commercial off-the-shelf components that are at risk due to vendor obsolescence, legislative changes, geopolitical factors, or design flaws. Track these items and coordinate multi-site Nuclear Security Enterprise (NSE) efforts to develop and qualify alternative material substitutes before production continuity is impacted.

What You'll Do Cont.

Project-Based & Advanced Engineering Duties

Design of Experiments (DoE) for Material Research: Plan, design, and execute rigorous experiments to characterize raw materials and evaluate long-term weapon compatibility.

Engineering Property Assessment and Manufacturing Optimization: Correctly and efficiently determine critical engineering properties (such as elastic modulus, thermal conductivity, density, or tensile strength). Apply materials science knowledge to propose process or material changes to existing weapon programs that optimize performance, reduce cost, and improve manufacturing yield.

New Process and Equipment Implementation: Lead and manage modernization projects, such as standing up new mixing, blending, and packaging equipment or transitioning legacy manual steps to advanced processes. Write and qualify the standard operating procedures needed to operate these new systems safely.

Conduct of Operations (ConOps) and Safety Basis Compliance: Participate in Management Self-Assessments (MSAs), readiness reviews, and hazard analyses before introducing new materials or process changes into nuclear explosive areas. Ensure that all material handling, storage (including high explosive storage limitations), and application processes strictly conform to Documented Safety Analyses (DSA) and federal directives.

Minimum Job Requirements

Bachelor's Degree in engineering/science/information technology discipline.Master's Degree in engineering/science/information technology discipline.

Department of Energy (DOE) Order 426.2A Requirements

In addition to the above requirements, the applicant must also meet the requirements of the Department Of Energy (DOE) O 426.2A (i.e.,Baccalaureate in engineering or related science, 2 years job-related and at least 1 year nuclear experience, or a DOE O 426.2A alternative). DOE O 426.2A experience requirements can be satisfied following employment. In the interim, newly hired personnel will not be assigned duties that could impact the safety basis of nuclear facilities. (Ref. DOE O 426.2A)

Preferred Job Requirements

A minimum Grade Point Average (GPA) is preferred for up to four years post-graduation, confirmed by transcript at the time of offer. Beyond four years from graduation, relevant experience will supersede GPA in hiring considerationsBachelor’s Degree with a minimum 3.0/4.0 cumulative GPAMaster’s Degree with a minimum 3.3/4.0 cumulative GPAPhD with a minimum 3.5/4.0 cumulative GPAGeneral knowledge of materials engineering and testing, kitting, and materials handlingKnowledge of procedure writing and a general understanding of the Unreviewed Safety Question (USQ) ProcessExperience working the Design Agencies and familiarity with common documentation requirements with the Design AgenciesKnowledge of System Applications and Products in Data Processing (SAP); demonstrated computer skills and knowledge of software applications including Microsoft Office; and the ability to effectively communicate through the oral and written word

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At Pantex, you won't just find a job—you’ll discover a career of purpose safeguarding national security through work performed nowhere else on Earth. We empower dedicated, innovative individuals like you to achieve their greatest impact surrounded by exceptional talent and limitless opportunities for professional growth.

When you #JointheMission, you choose a lifetime career where your commitment is genuinely valued and rewarded. At Pantex, we believe that extraordinary talent thrives when supported by a balanced life. Discover the flexibility that empowers you to excel, coupled with a benefits package designed for your total peace of mind from comprehensive health coverage and robust retirement planning, to opportunities for continuous learning through education reimbursement.

Notes

The minimum education and experience for the lowest career level in the job posting range are listed under Minimum Job Requirements. Successful candidates hired into a higher career level than the minimum in the range must meet the requirements listed in the job leveling charts for the career level into which they are being hired.

If a range of Career Levels is posted, i.e., Senior Associate to Senior Specialist, internal applicants already in one of the Career Levels would come across at their current Career Level. Internal applicants currently in a lower level Career Level would move to the lowest posted Career Level.

Requires a Q clearance; however all qualified candidates will be considered regardless of their current clearance status. The ability to obtain and maintain a Department of Energy Q clearance is required.

Position may require entry into Materials Access Areas (MAA) and participation in the Human Reliability Program (HRP). If HRP is required, candidate must complete a counterintelligence-scope polygraph, pursuant to 10 Code of Federal Regulations (CFR) 709. Medical requirements may apply.

Pantex is a drug-free workplace. Candidates accepting a job offer will be required to pass a pre-placement physical, drug screening and background investigation. As an employee, you may be required to receive and maintain a security clearance from the United States Department of Energy in order to meet eligibility requirements for access to sensitive information or matter. U.S. citizenship is a requirement for security clearance applicants. All employees are subject to random selection for drug testing without advance notification.

Pantex is an equal opportunity employer, and does not tolerate discrimination in the terms and conditions of employment on the basis of race, color, sex, sexual orientation, pregnancy, age, religion, national origin, ancestry, genetic information, disability, veteran status, or other characteristic protected by law. Such terms and conditions of employment include, but are not limited to, hiring, training, promotion, discipline, compensation, benefits, and termination of employment. Pantex

takes affirmative action to employ and advance in employment qualified individuals with disabilities and protected veterans; including disabled veterans, recently separated veterans, active-duty wartime, or campaign badge veterans, and Armed Forces service medal veterans. All employment decisions are based on business needs, job requirements, and individual qualifications.